ENGINEERING PROJECT REVIEW

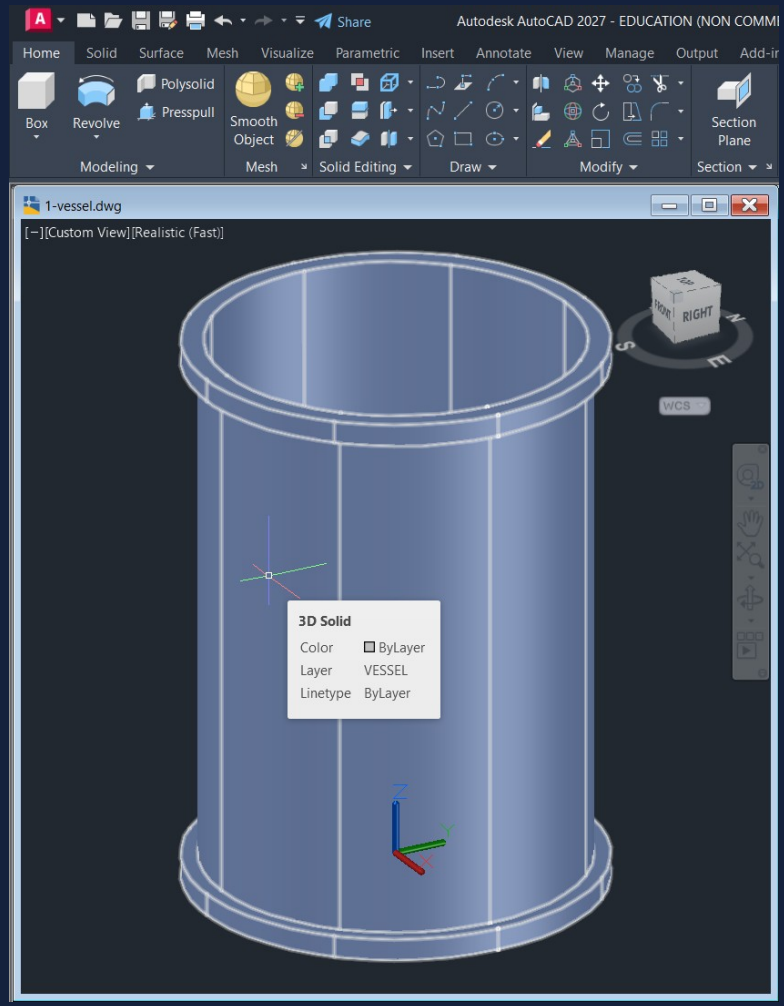
Crystal growth furnace setup

Part 1 — Design & Engineering Development (AutoCAD .dwg)

Part 2 — Structural & Buckling Analysis (Fusion → STEP → PrePoMax / CalculiX)

Part 3 — Steady-state conduction + cavity radiation, then coupled thermal expansion — 2D axisymmetric model in PrePoMax / CalculiX

Part 4 — Furnace heat balance and controls its temperature in C++



Project Workflow

One geometry, carried from 2D drafting to a verified finite-element model

1

AutoCAD

`.dwg`

2D layout, section & dimensions of vessel and internals



2

Autodesk Fusion

`.f3d, .step`

Import DWG sketch, build 3D solids, add ports, split to half model, for the FE pre-processor



3

PrePoMax

`.pmx`

Mesh, material specification, fixed + symmetry BCs, pressure loads



4

CalculiX

`.frd`

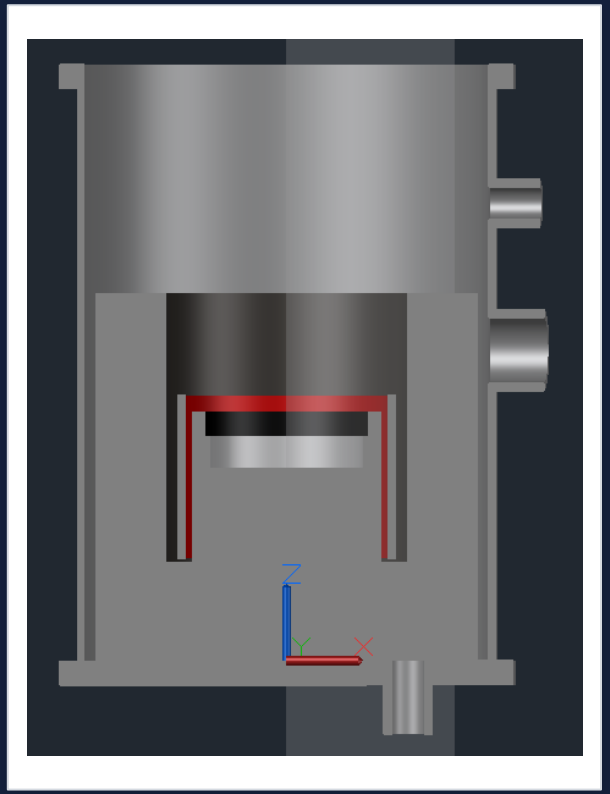
Static stress solve and linear buckling (eigenvalue) solve

CalculiX

PART 1

Pressure Vessel Design & Engineering Development

Design basis, geometry drafted in AutoCAD (.dwg), thickness sizing and preparation of the 3D model



Design Basis

Service conditions and requirements that drive the geometry

Parameter	Value
Internal design pressure	1.0 MPa (≈10 bar g)
External pressure	0.101 MPa — full vacuum inside, atmosphere outside
Material	Stainless steel AISI 304 (SS304)
Configuration	Vertical cylinder, flat bottom plate, top flange
Internals	Refractory insulation, crucible holder, crucible, charge, coil
Ports	Side viewport, side pyrometer port, bottom vacuum-pump port

A **Viewport**
Visual observation of the melt — largest side opening

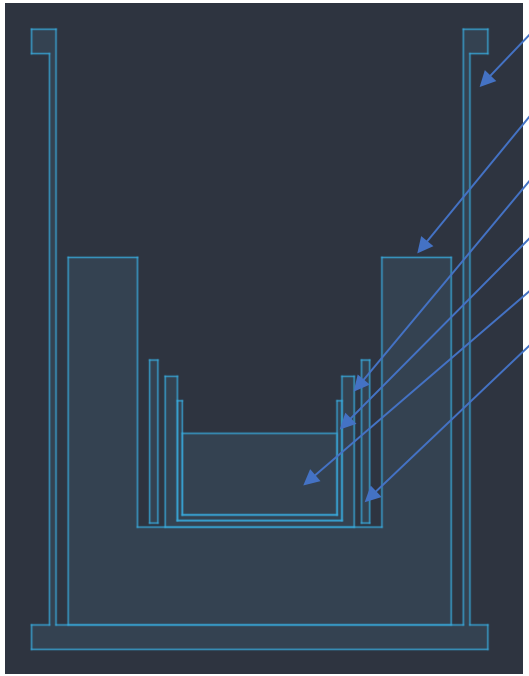
B **Pyrometer**
Non-contact temperature measurement

C **Vacuum pump**
Bottom port for evacuation to full vacuum

Two governing load cases follow from this basis: internal pressure (strength) and external pressure under vacuum (stability).

AutoCAD Layout — Vessel Cross-Section

Section through the vessel axis



- Drawn in 2D as a half-profile about the vessel axis, so every body can later be revolved into a solid.
- The vessel is one continuous body: 8 mm shell, 30 mm top flange and 30 mm flat bottom plate.
- Internals are nested concentrically: insulation → coil → holder → crucible → charge.
- Clearances between parts are kept so each body can be selected and meshed separately.

Key Dimensions

Measured from the drawing / exported STEP geometry (mm)

Ø500
Shell inside diameter

8
Shell wall thickness

700
Shell height

Ø560
Flange outside diameter

30
Flange & bottom plate

Component	Radius range	Height range (z)	Volume
Vessel (shell + flanges + bottom)	r 250 – 280	−30 → 730	17.8 L
Refractory insulation	r 0 – 235 (bore r 150)	0 → 450	54.7 L
Induction coil (ring)	r 125 – 135	125 → 325	1.6 L
Crucible holder	r 0 – 116	120 → 305	2.1 L
Crucible	r 0 – 101	128 → 275	0.7 L
Charge	r 0 – 95	135 → 235	2.8 L

Wall & Head Thickness Sizing

Indicative checks with ASME VIII-1 formulas — S = 138 MPa (SS304), joint efficiency E = 0.85

Cylindrical shell — UG-27

$t = P \cdot R / (S \cdot E - 0.6P)$

1 × 250 / (138 × 0.85 − 0.6) ≈ 2.1 mm

Selected 8 mm → large margin; thickness is driven by vacuum stability and stiffness at the ports.

Flat bottom head — UG-34

$t = d \cdot \sqrt{(C \cdot P / S \cdot E)}$

d = 500 mm, C = 0.17 – 0.33 → 18 – 25 mm

Selected 30 mm plate. Flat heads bend, so the plate centre is the expected location of peak deflection.

External pressure — vacuum

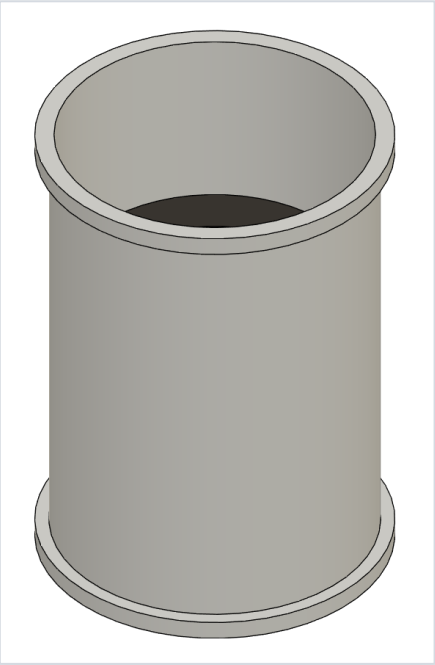
P = 0.101 MPa outside

Checked by linear buckling FEA in Part 2

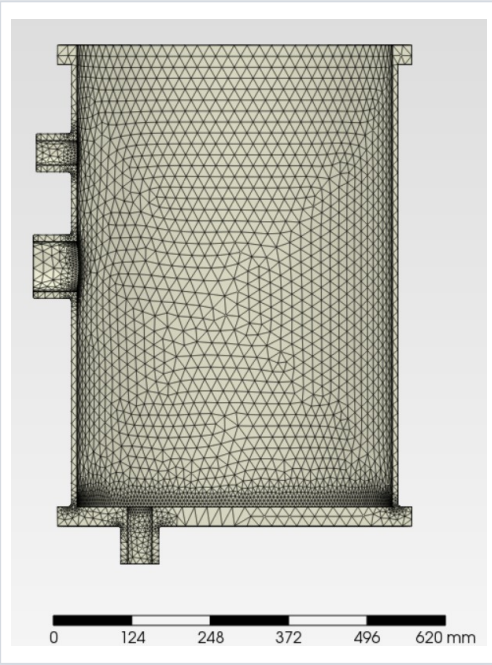
Thin shell (D/t ≈ 65) makes buckling the key stability criterion for the 8 mm wall.

From AutoCAD 2D to a 3D Half Model

DWG imported into Fusion 360, solids built, then exported as STEP



Fusion 360 — vessel body



Half model with ports (inside)

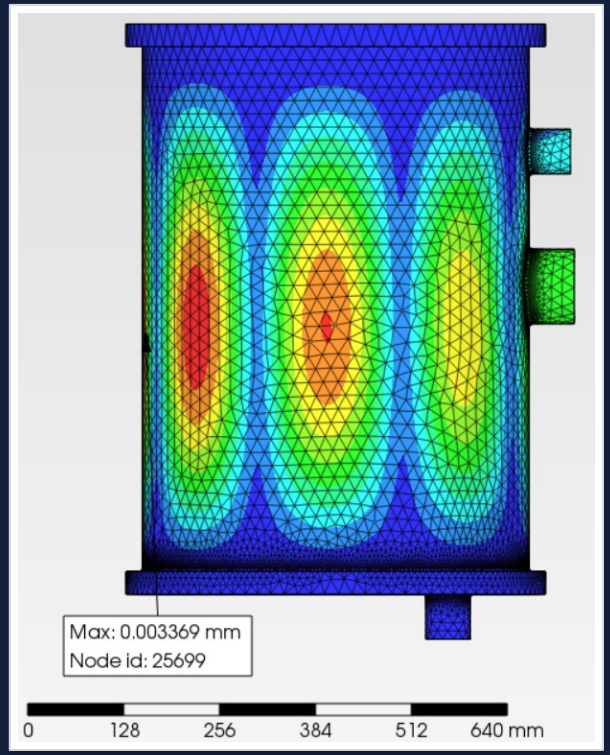
1. Import Semua_bagian.dwg into a Fusion sketch
2. Revolve profiles into solids; add viewport, pyrometer and pump ports
3. Split body along the symmetry plane → half model
4. Export Vesel-ported-8mm-half.step for PrePoMax

Why half? The geometry is symmetric about the cutting plane, so a symmetry BC halves the element count with no loss of accuracy.

PART 2

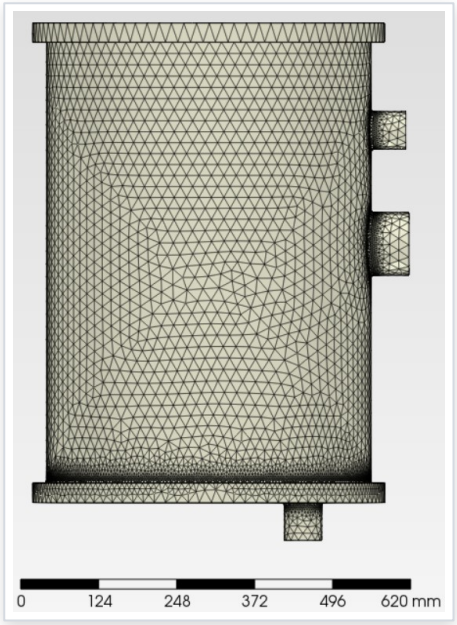
Pressure Vessel Structural & Buckling Analysis

STEP half model analysed in PrePoMax with the CalculiX solver: static pressure case and linear buckling under vacuum

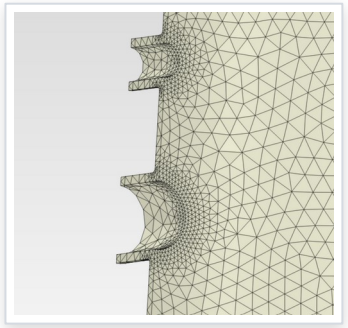


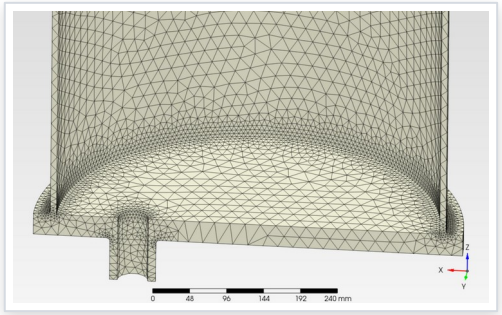
FE Model & Mesh

Second-order tetrahedral mesh with local refinement at the junctions



Global mesh (outside)





Local refinement: ports & wall–bottom joint

99,763

nodes

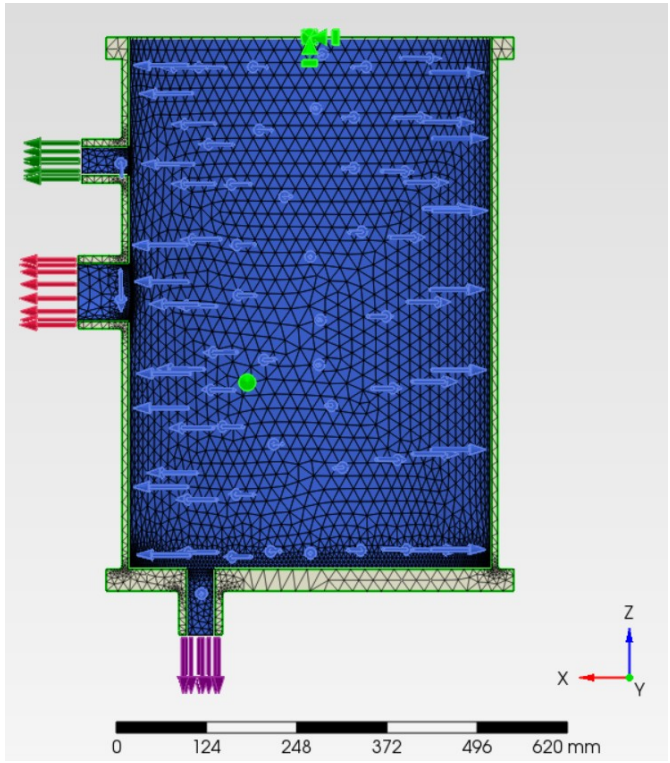
58,249

elements (2nd order)

- Material SS304: E = 193,000 MPa, ν = 0.29, ρ = 8×10^{-9} t/mm³.
- Local mesh size at the inner bottom corner and around each port, where stress gradients are highest.
- Node count sits between convergence runs of ~74k and ~108k nodes in the reference study (~119k), so results are expected within the converged band.

Static Case — Boundary Conditions & Loads

Internal pressure with equivalent cap loads on the port ends



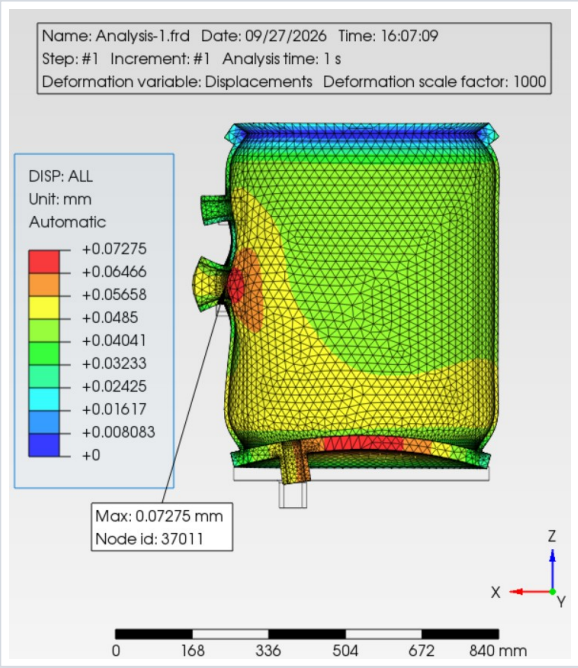
Item	Type	Value
Inner wetted surfaces	Pressure (outward)	1.0 MPa
Viewport cap	Pressure (pull-out)	−1.78 MPa
Pyrometer cap	Pressure (pull-out)	−0.80 MPa
Pump port cap	Pressure (pull-out)	−0.80 MPa
Top flange	Fixed support	u = 0
Cut plane	Symmetry	u normal = 0

Cap loads represent the blind-flange force transmitted by each port: pressure on the closed area divided by the port-end ring area, applied with negative sign so it pulls the port away from the vessel.

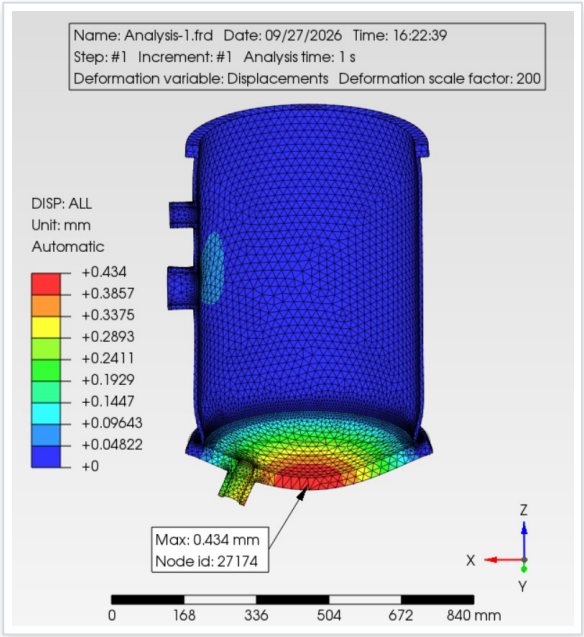
Blue = internal pressure · red/green/purple = port caps · green = fixed & symmetry

Verification: Catching a Missing Load

The first run solved without errors, but the numbers were wrong



Run 1 — max 0.073 mm, bottom moves up



Corrected — max 0.434 mm at plate centre

Symptom

Max deflection was 5–6× lower than the reference, and the bottom plate lay above its undeformed outline.

Diagnosis

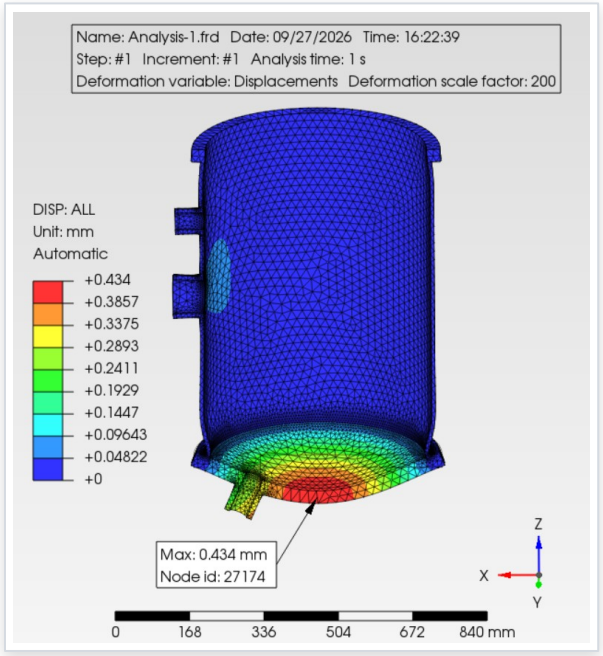
Shell growth (≈0.04 mm) matched hand calcs, so material and load magnitude were right; the internal pressure simply was not applied to the bottom floor.

Fix

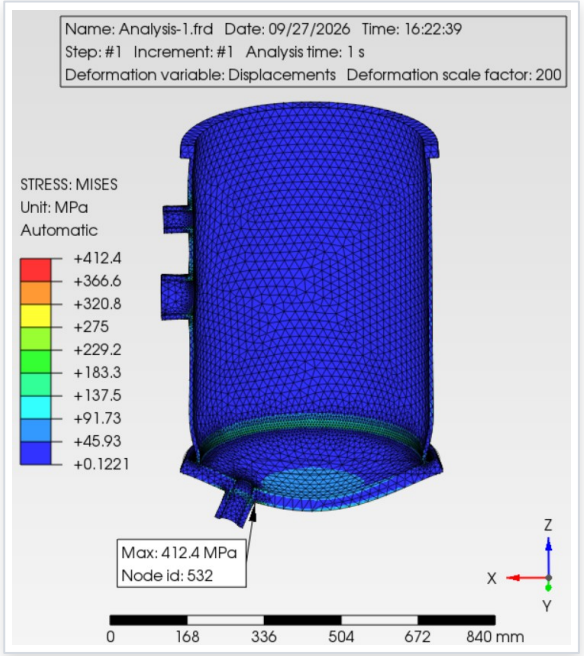
Floor and inner fillet added to the pressure surface set, then re-run.

Static Results — Deformation, Stress & Safety Factor

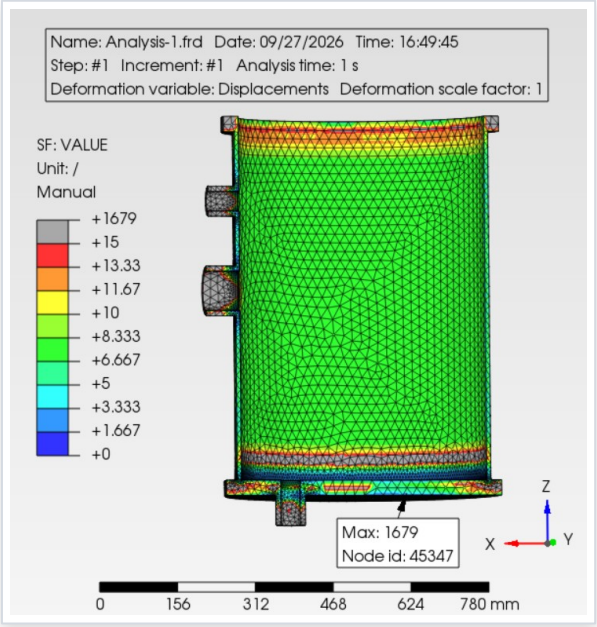
1 MPa internal pressure, SS304 linear elastic



Displacement (×200)



Von Mises stress (×200)



Safety factor (scale capped at 15)

0.434 mm max displacement, bottom-plate centre

412.4 MPa peak von Mises, bottom-port junction

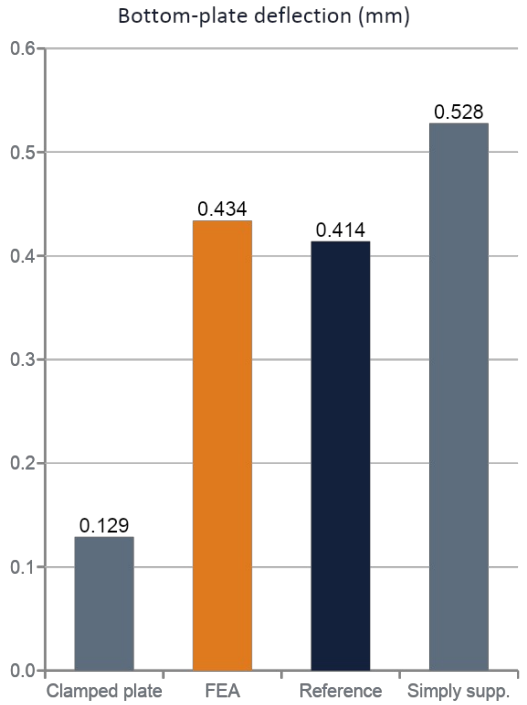
≈ 31 MPa membrane hoop stress in shell

Hand Calculations vs FEA

Independent checks confirm the model behaves as expected

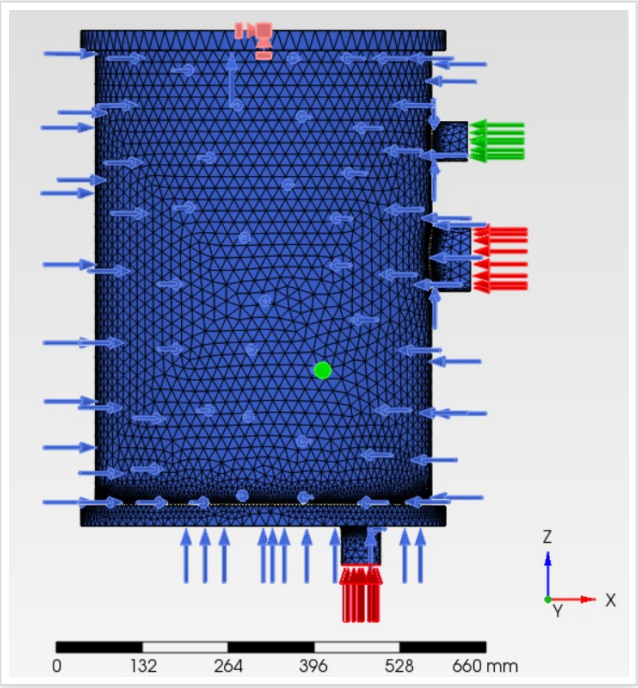
Quantity	Hand calculation	FEA	Verdict
Hoop stress, shell (pR/t)	31.3 MPa	0 – 46 MPa band	Agrees
Radial growth, shell	0.036 mm	≈ 0.04 mm	Agrees
Bottom-plate centre deflection	0.13 (clamped) – 0.53 mm (simply supp.)	0.434 mm	Within bounds
Reference study deflection	0.414 mm	0.434 mm	+4.8 %
Pre-buckling deflection (0.434 × 0.101)	0.0438 mm	0.0436 mm	Agrees

The flat bottom behaves between the clamped and simply-supported limits — the flange-to-plate joint gives partial rotational restraint.

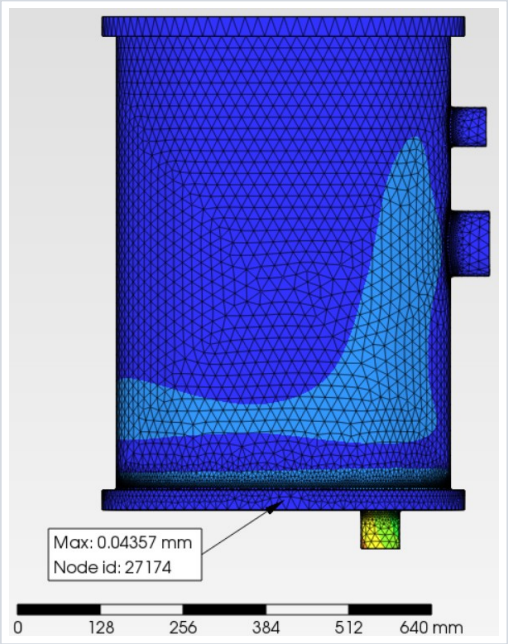


Buckling Case — Vacuum Loading

Linear (eigenvalue) buckling in CalculiX, same mesh and supports



External pressure and port loads



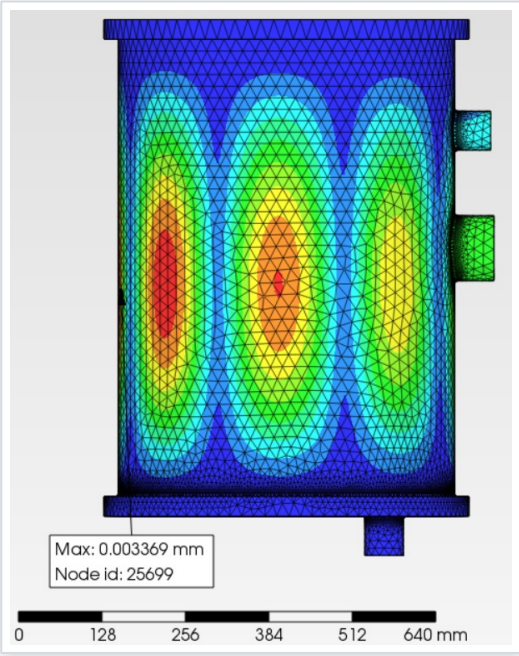
Pre-buckling state, max 0.0436 mm

Load	Value
External pressure (all outer faces)	0.101 MPa
Port cap load A	0.281 MPa
Port cap load B	0.182 MPa
Supports	Fixed flange + symmetry

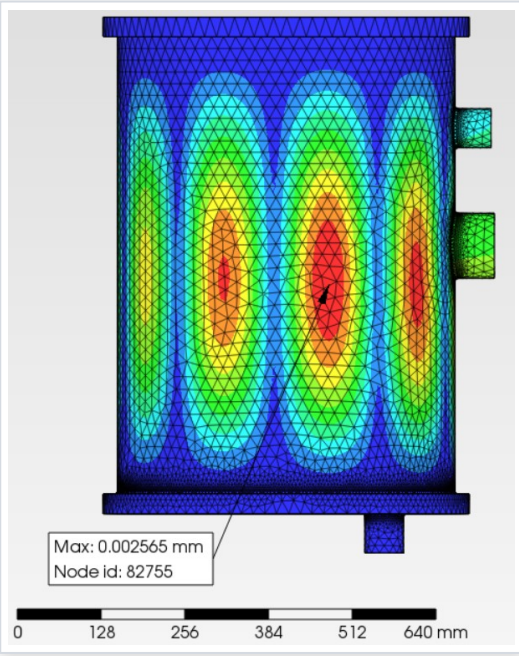
The solver returns load multipliers (buckling factors): the critical load equals the factor times the applied vacuum load set.

Buckling Mode Shapes

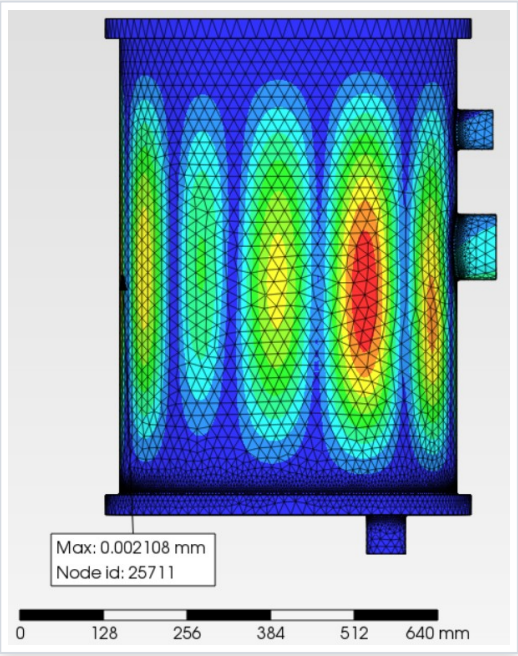
First three eigenmodes — circumferential lobes in the cylindrical shell



Mode 1 $\lambda = 144.8$



Mode 2 $\lambda = 165.2$

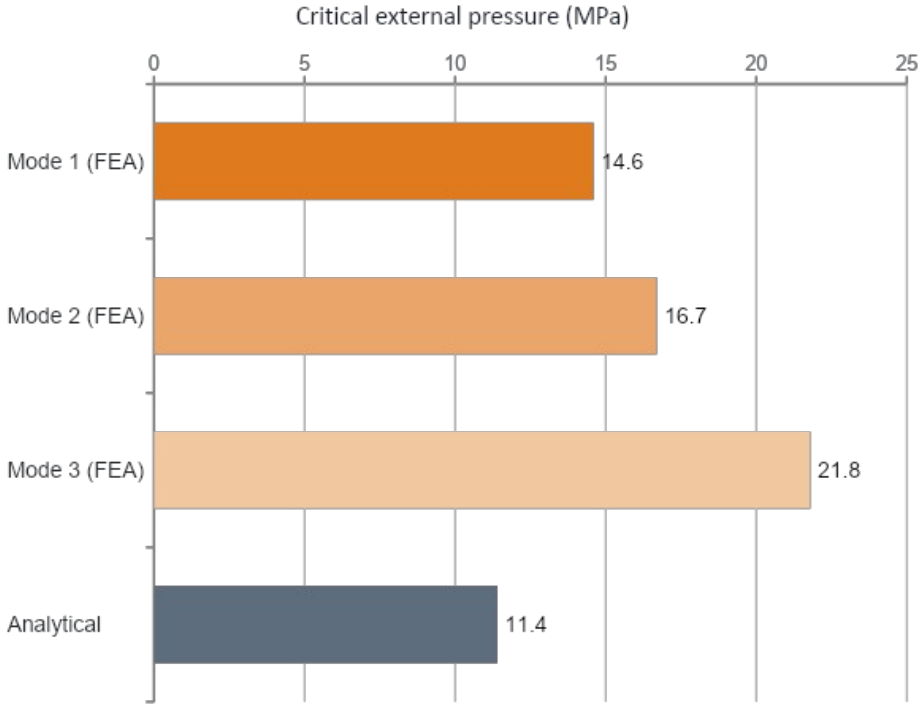


Mode 3 $\lambda = 215.9$

Lobes form between the stiff flange and bottom plate; the ports locally perturb the pattern.

Buckling Interpretation

Critical external pressure versus the 0.101 MPa vacuum load



144.8

lowest buckling factor (Mode 1) → Pcr ≈ 14.6 MPa

- Analytical short-cylinder estimate (Windenburg–Trilling, L/D = 1.36, D/t = 65) gives ≈ 11.4 MPa; FEA is higher because the flange and 30 mm bottom stiffen the ends.
- Even with a typical knock-down of 0.2 for imperfections, the margin is ≈ 29× over full vacuum — far above the usual minimum of 3.
- Buckling does not govern the 8 mm wall; there is room to study thinner walls against strength and stiffness.

Conclusions & Next Steps

Findings

- AutoCAD → Fusion → STEP → PrePoMax workflow produced a consistent half model.
- Static deflection 0.434 mm, within 4.8 % of the reference and bounded by plate theory.
- Shell membrane stress ≈ 31 MPa, well below SS304 yield (≈ 205 MPa).
- Lowest buckling factor 144.8 under full vacuum — stability is not critical.

Open issues

- Peak 412.4 MPa at the bottom-port junction exceeds yield locally; it sits at a sharp re-entrant corner and is mesh-sensitive.
- Linear buckling ignores imperfections and plasticity.

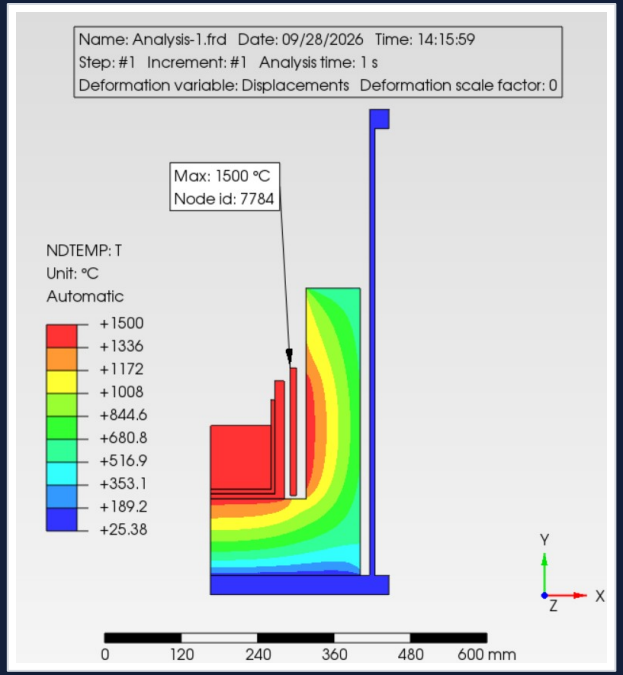
Next steps

- Add a fillet / reinforcing pad at the bottom port and re-run.
- Linearised stresses through the thickness (ASME VIII-2 Part 5) to separate membrane, bending and peak.
- Mesh-convergence check at the hotspot; compare wall-thickness variants.

PART 3

Hot-Zone Thermal & Thermo-Mechanical Analysis

Steady-state conduction + cavity radiation, then coupled thermal expansion — 2D axisymmetric CAX6 model in PrePoMax / CalculiX 2.22



Part 3 Workflow

From the half cross-section to temperature and deformation fields

1

Fusion 360

Half r–z faces, rotated to XY, normals +Z → STEP

2

PrePoMax

2D axisymmetric, 6 parts, 2nd-order CAX6 mesh

3

Thermal step

Fixed T, water/air films, cavity radiation

4

Coupled step

Uncoupled temperature–displacement, contact, gravity

5

Post-process

Contours, point queries, hand-calc checks

Axisymmetry: X = radius r ≥ 0, Y = vessel axis. CalculiX expands each 2D element internally into a thin 3D wedge.

FE Model & Mesh

Six separate parts joined by ties; second-order triangles refined in thin walls



Second-order mesh, all parts

22,954

nodes

10,716

CAX6 elements

5 / 0.5 mm

max / min element size

Part	Material	Local mesh
Vessel	SS304	5 mm
Insulator	Graphite felt	5 mm
Susceptor (holder)	Graphite IG-110	2 mm
Heater coil	Graphite IG-110	2 mm
Crucible	Fused silica	1.5 mm
Si melt	Liquid Si	5 mm

Parts were kept separate (no compound) so each can carry its own material section.

Material Properties

Temperature-dependent conductivity; mechanical data only for load-carrying parts

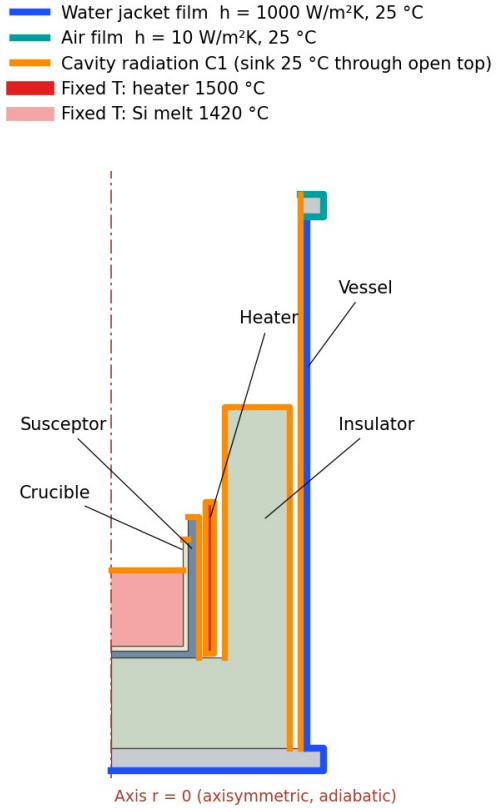
Material	k [W/m·K] 20 → 1500 °C	ε	E [GPa] / ν	α [10^{-6}/K]	ρ [kg/m³]
SS304 (vessel)	16.2 → 28	0.35	193 / 0.29	17.3	8000
Graphite felt (insulator)	0.08 → 0.55	0.85	0.001* / 0.3	—	100
Graphite IG-110 (susceptor, heater)	120 → 50	0.85	9.8 / 0.20	4.5	1770
Fused silica (crucible)	1.38 → 3.0	0.80	72 / 0.17	0.55	2200
Si melt	50 (T fixed)	0.30	0.001* / 0.3	—	2570

* **Dummy stiffness (E = 1 MPa).** Felt and melt carry no structural load but must stay in the mesh so the thermal and structural fields share the same nodes.

Zero-strain temperature 20 °C for all expansion data. Physical constants: absolute zero −273.15 °C, Stefan–Boltzmann 5.67×10^{-11} mW/mm²K⁴ (mm-t-s units).

Thermal Boundary Conditions

Steady-state heat transfer with surface-to-surface cavity radiation

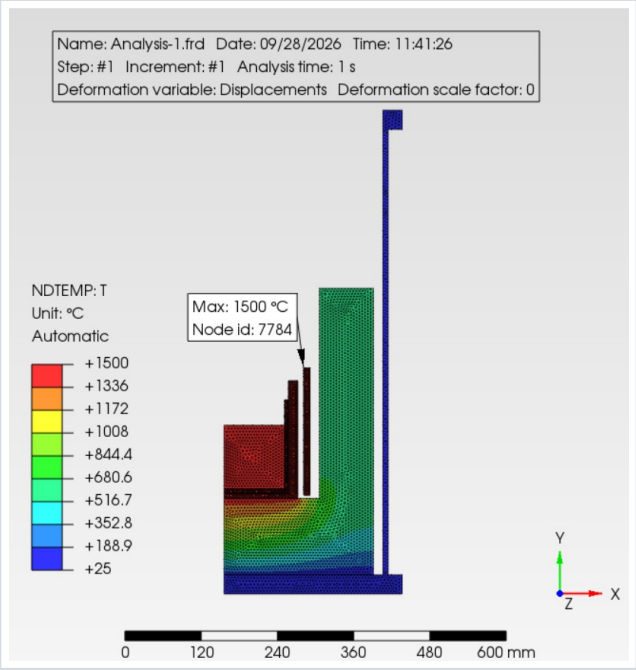


Item	Surfaces	Value
Fixed temperature	Heater coil / Si melt	1500 / 1420 °C
Water-jacket film	Shell outer wall + bottom plate	h = 1000 W/m²K, 25 °C
Air film	Top flange	h = 10 W/m²K, 25 °C
Cavity radiation C1	Heater, susceptor, crucible rim, melt top, insulator, vessel inner wall	ε = 0.85 / 0.85 / 0.80 / 0.30 / 0.85 / 0.35
Radiation sink	Open top (cold lid)	25 °C
Ties (4)	Vessel–insulator, insulator–susceptor, susceptor–crucible, crucible–melt	Perfect conduction

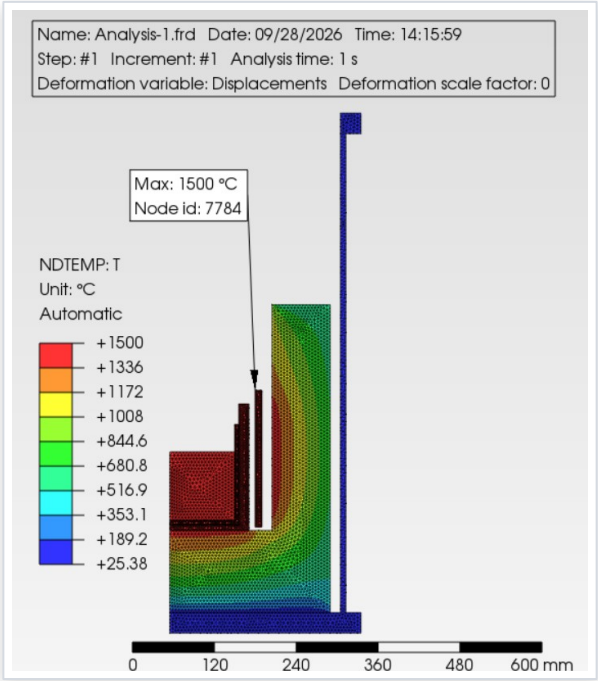
Edges already tied to a neighbour were excluded from radiation, so heat is not radiated through solid parts.

Why Radiation Matters

Conduction-only first run vs full model with cavity radiation



Run 1 — conduction only: heater isolated at 1500 °C

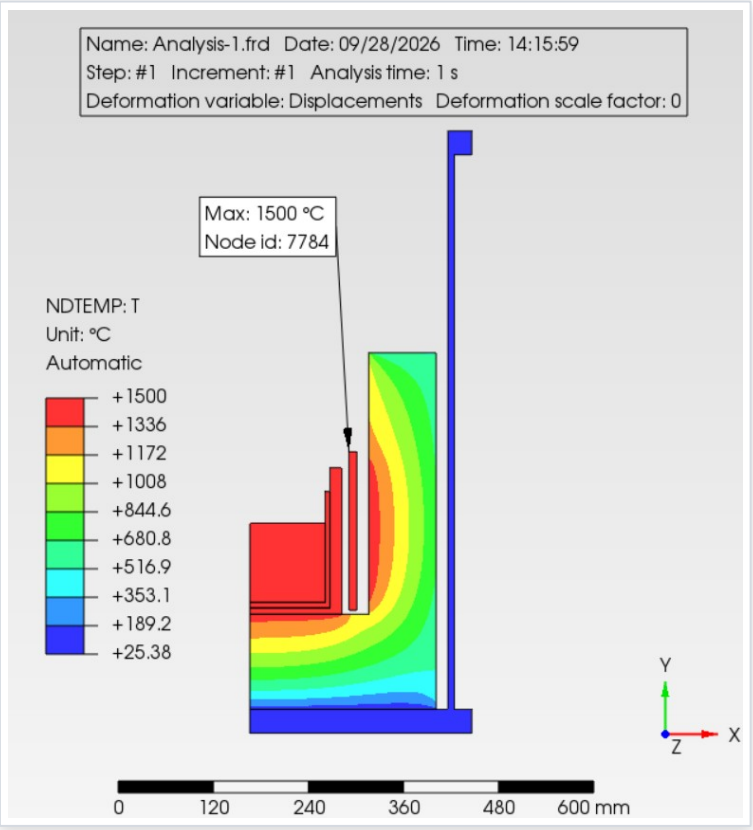


Run 2 — with cavity radiation C1

- The heater touches nothing, so without radiation it cannot heat the hot zone.
- With radiation the insulator bore wall heats up and a smooth gradient forms outward.
- Solve time rose from ~3 s to ~189 s because radiation is strongly nonlinear ($\propto T^4$).

Temperature Field — Full Assembly

Steady state, automatic legend 25–1500 °C



1500 °C

heater (prescribed)

1459 °C

susceptor outer wall, z = 203

707 °C

insulator outer wall, z = 230

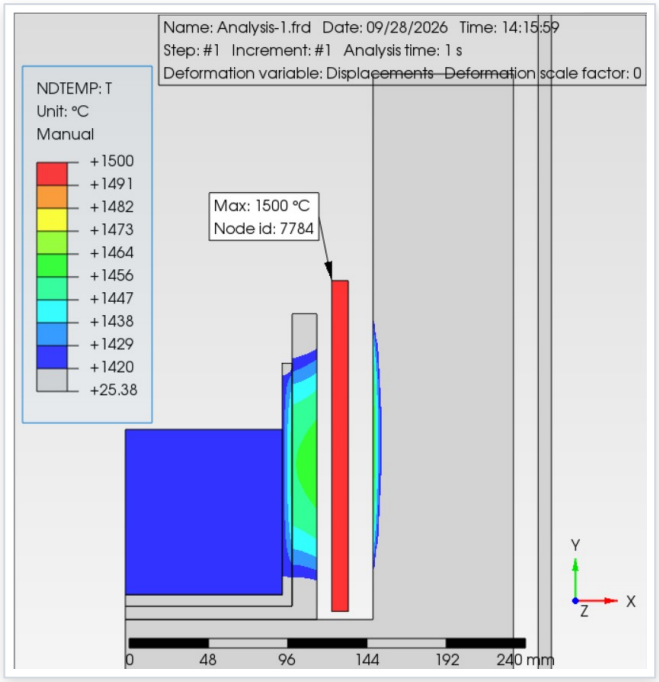
66.9 °C

vessel inner wall, z = 208

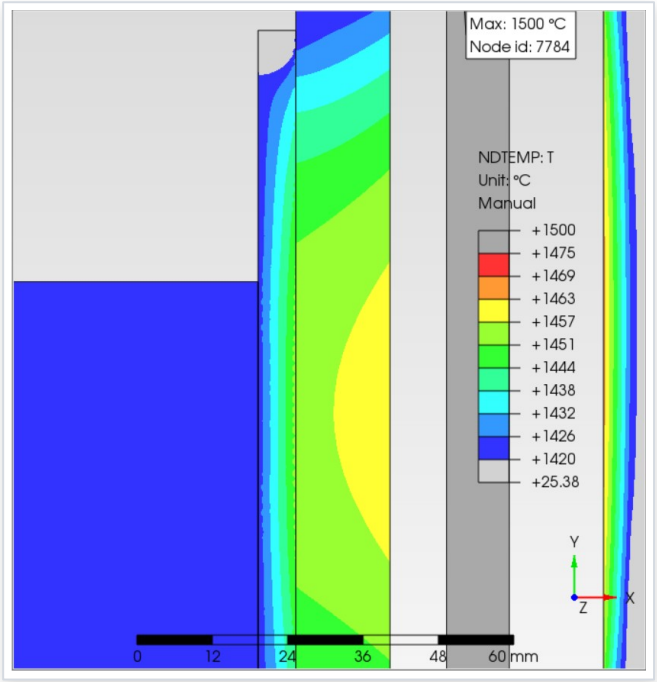
- Hot zone (melt, crucible, susceptor) sits between 1420 and 1500 °C.
- Insulation drops the temperature by ~700 K across 85 mm of felt.
- Water jacket keeps the vessel wall below 70 °C.

Hot-Zone Detail — Susceptor & Crucible

Manual legends isolate the 1420–1500 °C band (grey = outside range)



Susceptor & heater, 1420–1500 °C

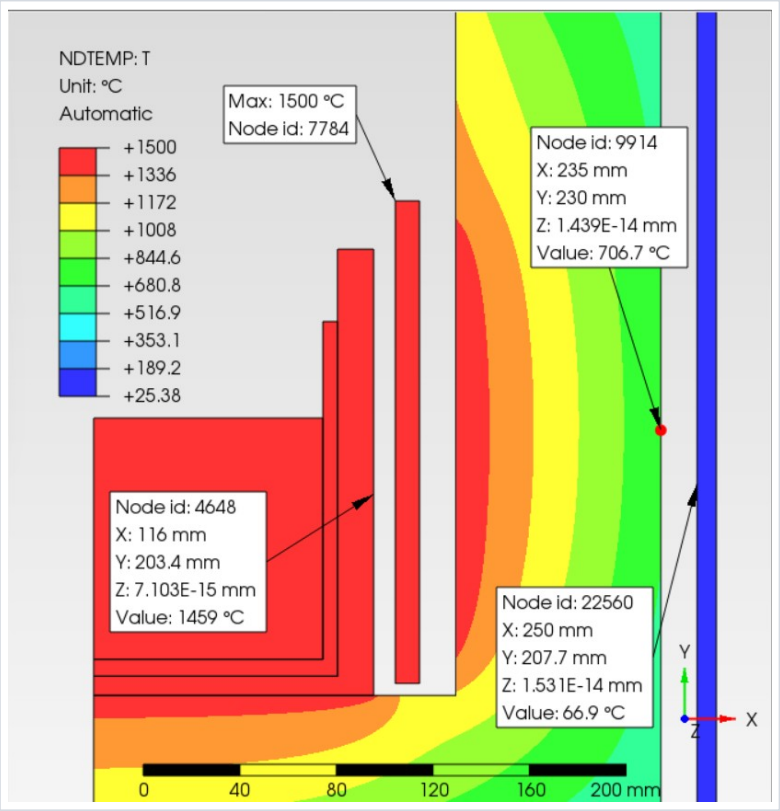


Crucible & susceptor wall, 1420–1475 °C

- Susceptor peaks mid-height on the heater-facing wall (~1457–1463 °C).
- Crucible wall stays close to the melt temperature (≈1420–1430 °C).
- Rims above the melt fall below 1420 °C: the insulator is open at the top and they radiate to the cold lid.

Vessel Temperature — Hand-Calculation Check

Point queries at z ≈ 200–230 mm compared with 1D wall conduction

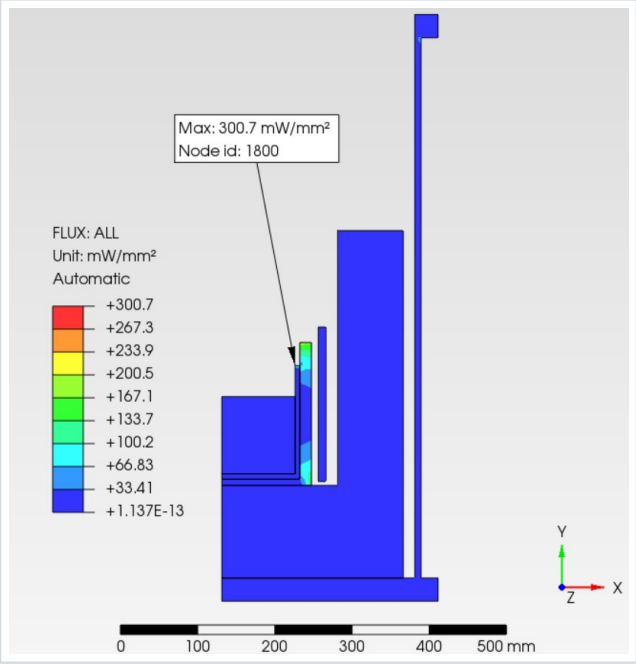


Step	Relation	Result
Heat flux into wall (FE query)	—	29.25 kW/m²
Water film ΔT	q / h = 29 250 / 1000	≈ 29 K → outer wall ≈ 54 °C
8 mm SS304 wall ΔT	q·t / k = 29 250 × 0.008 / 16	≈ 14 K → inner wall ≈ 68 °C
FE inner wall	Query node 22560	66.9 °C ✓

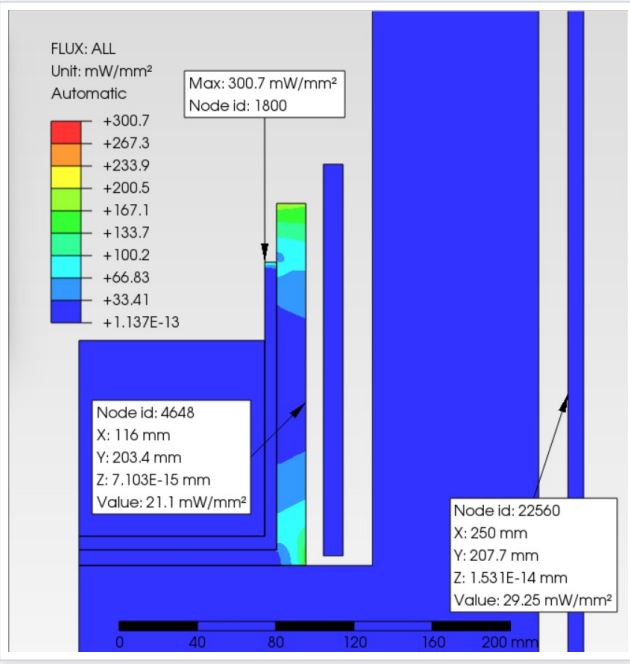
The vessel response to the heat it receives is computed correctly. The open question is how much heat reaches it — see the radiation balance check.

Heat Flux Distribution

Magnitude of the conductive heat flux (1 mW/mm² = 1 kW/m²)



Full model



Hot zone with point queries

21.1 kW/m²

susceptor outer wall (ref. 20.2 kW/m² into melt)

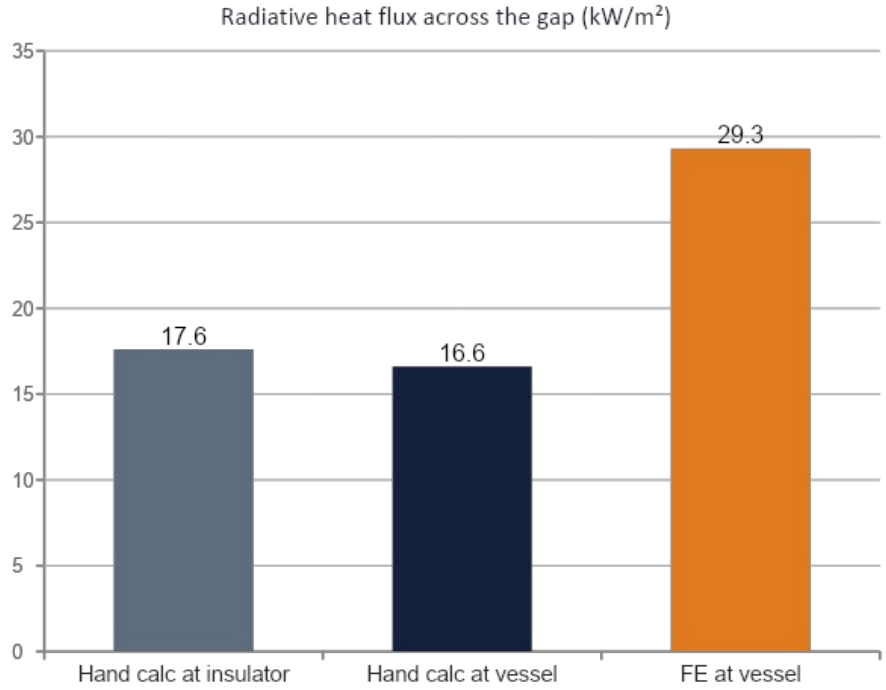
29.3 kW/m²

vessel inner wall

- The 300.7 mW/mm² maximum is a single node at the crucible-rim corner where the tie ends — a geometric singularity, not a design value.
- A manual 0–50 mW/mm² legend is recommended for the report figure.

Verification — Radiation Balance

Insulator outer wall (707 °C, ε 0.85) facing vessel inner wall (67 °C, ε 0.35)

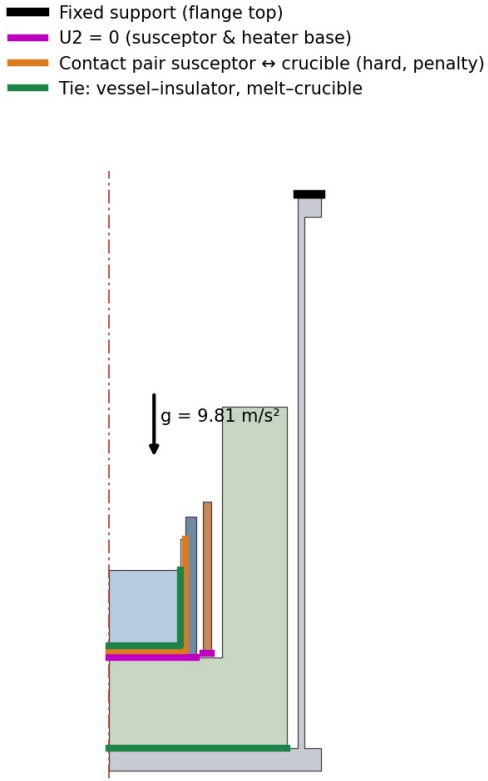


$$q = \sigma(T_1^4 - T_2^4) \,/\, [1/\varepsilon_1 + (r_1/r_2)(1/\varepsilon_2 - 1)]$$

- Two long concentric cylinders, r_1 = 235 mm, r_2 = 250 mm: 51.5 / 2.92 ≈ 17.6 kW/m².
- FE flux at the vessel is ~1.8× the hand estimate.
- Emissivities, cavity name, sink and selected edges were all re-checked and are correct.
- Likely cause: view-factor accuracy of CalculiX cavity radiation in the 2D axisymmetric expansion (to be confirmed).
- Effect: vessel temperature is over-estimated — conservative for the vessel.

Thermo-Mechanical Approach

Temperature and deformation solved together in one step



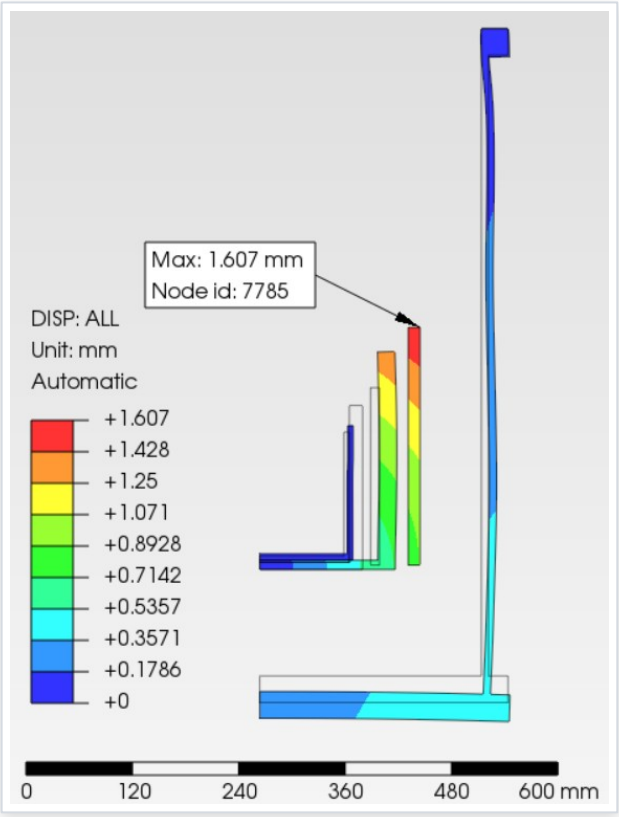
Lesson learned

Importing temperatures from the thermal .frd (*TEMPERATURE, FILE) left every node at 20 °C: the file only holds the 2D nodes, while CalculiX evaluates expansion on the internally expanded 3D nodes. Max displacement was only 0.096 mm.

- Fix: Uncoupled temperature–displacement step (steady state) with all thermal loads re-applied.
- Initial temperature 20 °C = stress-free state.
- Tie susceptor–crucible replaced by hard (penalty) contact so the side gap can open.
- Supports: flange top fixed; susceptor and heater bases U2 = 0; gravity 9.81 m/s².
- Solve time 2216 s (automatic increments from 6.25 %).

Thermal Expansion — Displacement

Thermal + gravity, structural bodies only (felt and melt hidden)



|U| [mm], deformed shape exaggerated

Part	FE |U|	Hand check	Reference
Heater (max, top)	1.607 mm	α·ΔT·(L, r) ≈ 1.6 mm	—
Susceptor top	≈ 1.3–1.4 mm	—	1.43–1.45 mm ✓
Crucible	≈ 0.1–0.3 mm	—	0.14 mm ✓
Vessel bottom plate	≈ 0.4–0.5 mm	17.3e-6 × 47 K × 730 mm ≈ 0.6 mm	≈ 0.1 mm

- CTE mismatch (graphite 4.5 vs silica 0.55 $\times 10^{-6}$/K) pulls the susceptor wall away from the crucible — the side gap opens, as in the reference.
- Vessel is fixed at the flange top, so its warmer wall (67 °C) grows downward; this is larger than the reference because the vessel runs hotter.

Comparison with Reference Study

Where the model agrees and where it differs

Quantity	This model	Reference	Assessment
Susceptor temperature	1459 °C (mid outer wall)	1434–1480 °C	Agrees
Heat flux at susceptor	21.1 kW/m²	20.2 kW/m² (into melt)	Agrees
Vessel wall temperature	66.9 °C	25–34 °C	Hotter — open-top insulator + radiation balance gap
Susceptor displacement	≈ 1.3–1.4 mm	1.43–1.45 mm	Agrees
Crucible displacement	≈ 0.1–0.3 mm	0.14 mm	Agrees
Vessel displacement	≈ 0.4–0.5 mm	≈ 0.1 mm	Larger — follows hotter vessel
Interface model	Tie (perfect contact)	h_c = 200–5000 W/m²K study	Sensitivity not covered

Takeaway: the hot zone and its thermal expansion agree with the reference. The differences all trace back to two causes — the open-top insulator geometry and the radiation balance at the vessel — and both push the vessel results to the conservative side.

Conclusions & Next Steps

Findings

- Hot-zone temperatures match the reference (susceptor 1459 °C).
- Felt insulation carries a ~700 K drop; vessel stays below 70 °C.
- Thermal expansion reproduces the reference trend: susceptor ≈ 1.4 mm, crucible ≈ 0.14 mm, side gap opens.

Limitations

- Insulator is open at the top (no felt lid) — rims run cooler, vessel hotter.
- Radiation balance: FE flux ~1.8× the hand estimate at the vessel.
- Ties give perfect thermal contact; felt/melt use dummy stiffness.

Next steps

- Combined load case: hydrostatic melt pressure + 0.101 MPa vacuum → stress & safety factor.
- Contact-layer model for h_c sensitivity.
- Line profiles and 3D revolved views (ParaView); mesh study of radiation view factors.

PART 4

Furnace Energy Model & Control Software in C++

Stage by stage — every step derived by hand, verified by a test, and recorded in Git



Control chain architecture

1

Recipe

Process sequence & setpoints: temperature ramps, targets, dwell times, gas conditions, growth parameters

2

Control Logic

Converts requirements into actions: sequencing, PID feedback, interlocks, alarms, safety conditions

3

I/O Interface

Acquires sensor data and sends commands to heaters, valves, flow controllers, power supplies

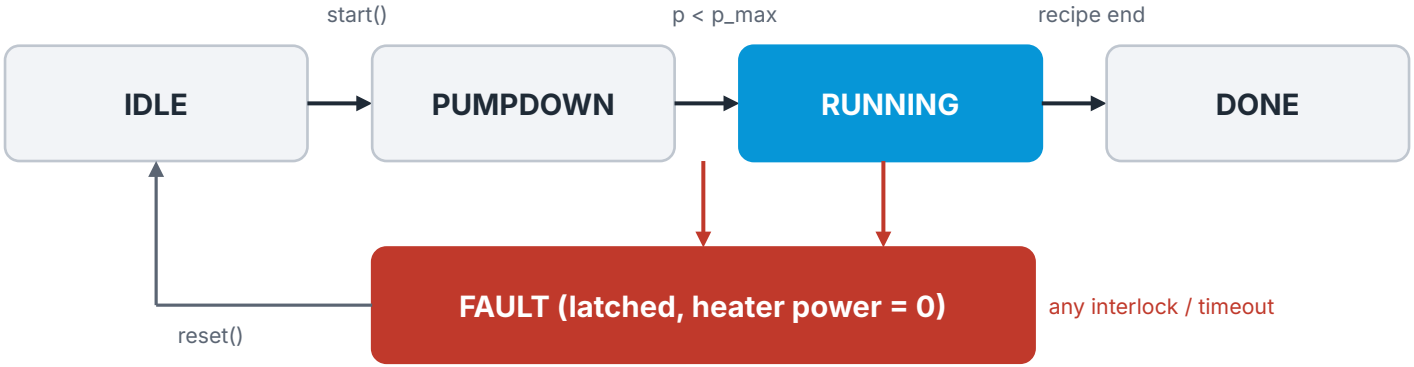
4

Physical / Simulated Furnace

Real process response, or a validated digital twin for development, testing and validation

Process state machine and safety interlocks

Faults are latched: heater off until the cause is cleared and an operator reset is given

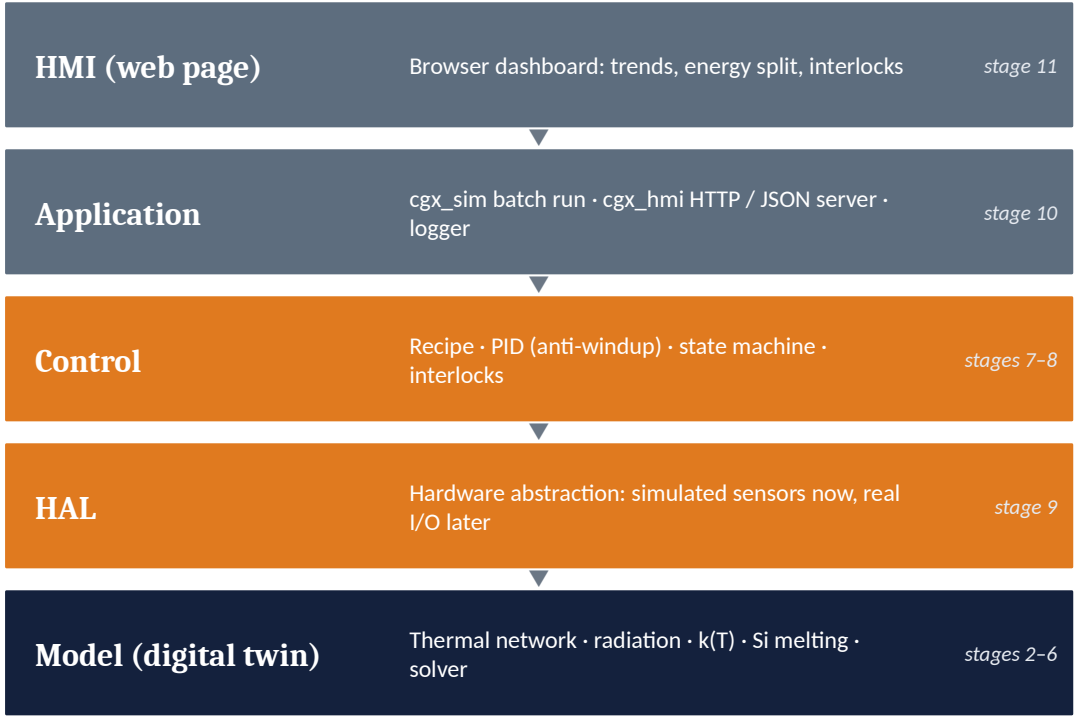


Interlock	Trips when
`cooling_water_flow`	flow < minimum or signal invalid
`cooling_water_temperature`	outlet > T_water_out_max
`susceptor_overtemperature`	control T > T_susceptor_max
`temperature_sensor`	pyrometer invalid while TC > hand-over + 50 K
`vacuum`	p > p_heating_max while RUNNING
`power_supply`	supply reports a fault

- **Temperature measurement**
- Type-C thermocouple at low T; pyrometer above the hand-over temperature (30 K hysteresis)
- **Control**
- PID heater power with anti-windup and power slew limiting
- **Safety**
- Software interlocks complement – never replace – the hardwired safety chain

Software Design — Layered Architecture

Each layer only talks to the layer below it



Why this structure

- The same control code runs on the simulator and on real hardware — only the HAL changes.
- The physics model can be tested alone against hand calculations and FEA.
- Every layer has its own unit tests, so a bug is found where it is made.
- Built bottom-up: model first, HMI last.

Development Roadmap — 13 Stages

One growing project; each stage ends with passing tests and a Git tag

0	1	2	3	4	5	6
Specification	Toolchain	Single RC node	Steady network + LU	Radiation + Newton	Furnace model vs FEA	Transient + melting

7	8	9	10	11	12
PID + recipe	States + interlocks	HAL + sensors	Logger + sim app	HMI web	Documentation

Done · In progress · Started · Planned

Stages 2–4 are the numerical foundation; stage 5 onward assembles them into the furnace and its controls.

Working Method — Proving Every Step

The reference project is used only as an answer key

1

Hand calculation

Derive the formula and the expected numbers on paper first.

2

Test

Write a test that compares the code with the hand result.

3

Code

Implement until the test passes.

4

Record

Notes in docs/stages, commit and Git tag per stage.

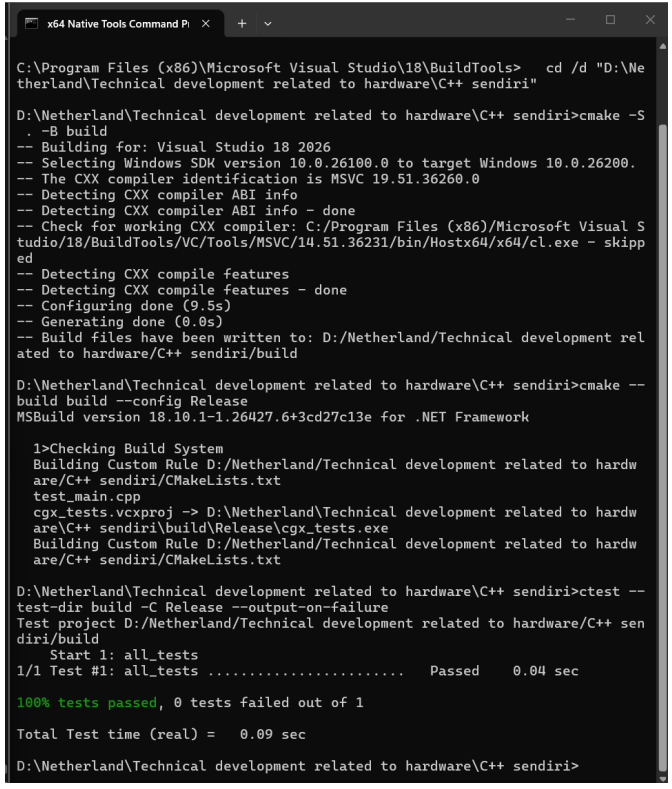
Repository layout

```
Cpp-crystal-growth/
├─ CMakeLists.txt   .gitignore
├─ include/  src/  tests/
├─ apps/  config/  web/
└─ docs/stages/00_specification.md … 12_documentation.md
```

- One project that grows — no copied code per stage.
- git tag stage-01, stage-02 … lets any stage be reopened.
- Every stage note ends with three questions answered in my own words.

Stage 1 — Toolchain (Done)

C++17 · CMake · CTest · Git · VS Code



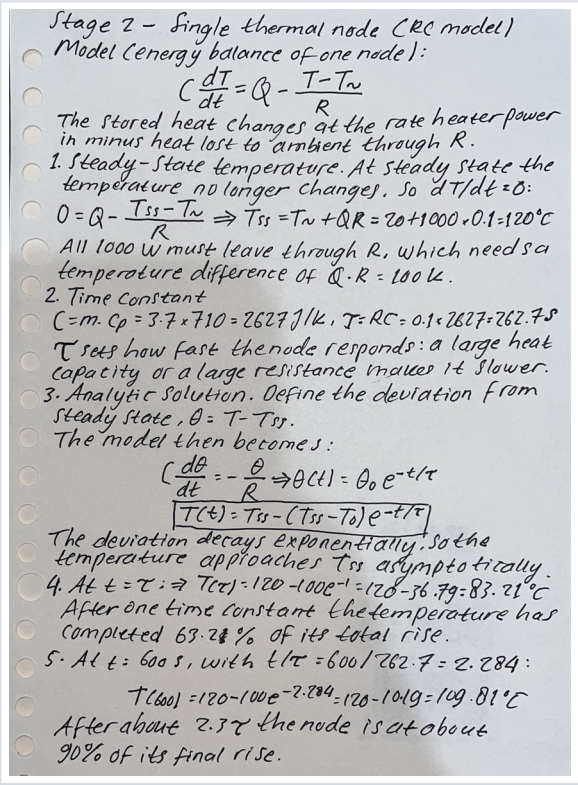
First build and test: 100 % passed

Item	Choice
Compiler	MSVC 19.51 (Build Tools for Visual Studio 2026)
Build system	CMake → Visual Studio 18 2026 generator
Tests	CTest; test program returns 0 = pass
Editor	Visual Studio Code
Version control	Git, tag stage-01

- CMake does not compile — it generates build files that call cl.exe.
- build/ is separate and ignored by Git; it can be deleted and rebuilt.
- Folder renamed to Cpp-crystal-growth (no spaces or “+”), build regenerated.

Stage 2 — Single Thermal Node (RC Model)

Smallest possible energy model: one heat capacity, one resistance



Stage 2 – Single thermal node (RC model)
Model (energy balance of one node):

$$C\frac{dT}{dt} = Q - \frac{T - T_\infty}{R}$$

The stored heat changes at the rate heater power in minus heat lost to ambient through R.

1. Steady-state temperature. At steady state the temperature no longer changes, so $dT/dt = 0$:

$$0 = Q - \frac{T_{ss} - T_\infty}{R} \Rightarrow T_{ss} = T_\infty + QR = 20 + 1000 \cdot 0.1 = 120\,°C$$

All 1000 W must leave through R, which needs a temperature difference of $Q \cdot R = 100$ K.

2. Time constant

$C = m \cdot c_p = 3.7 \times 710 = 2627$ J/K, $\tau = RC = 0.1 \times 2627 = 262.7$ s

τ sets how fast the node responds: a large heat capacity or a large resistance makes it slower.

3. Analytic solution. Define the deviation from steady state, $\theta = T - T_{ss}$.

The model then becomes:

$$C\frac{d\theta}{dt} = -\frac{\theta}{R} \Rightarrow \theta(t) = \theta_0 e^{-t/\tau}$$

$$\boxed{T(t) = T_{ss} - (T_{ss} - T_0)e^{-t/\tau}}$$

The deviation decays exponentially, so the temperature approaches T_{ss} asymptotically.

4. At $t = \tau$: $T(\tau) = 120 - 100e^{-1} = 120 - 36.79 = 83.21\,°C$

After one time constant the temperature has completed 63.2 % of its total rise.

5. At $t = 600$ s, with $t/\tau = 600/262.7 = 2.284$:

$$T(600) = 120 - 100e^{-2.284} = 120 - 10.19 = 109.81\,°C$$

After about 2.3τ the node is at about 90% of its final rise.

$$C \cdot dT/dt = Q - (T - T\infty) / R$$

Symbol	Value	Source
m	3.7 kg	susceptor 2.1 L × 1.77 g/cm³ (Part 1)
c_p	710 J/kg·K	graphite, room temperature
C = m·c_p	2627 J/K	
R	0.1 K/W	chosen
Q	1000 W	heater
T∞ = T₀	20 °C	ambient / start

120 °C

steady state T∞ + Q·R

262.7 s

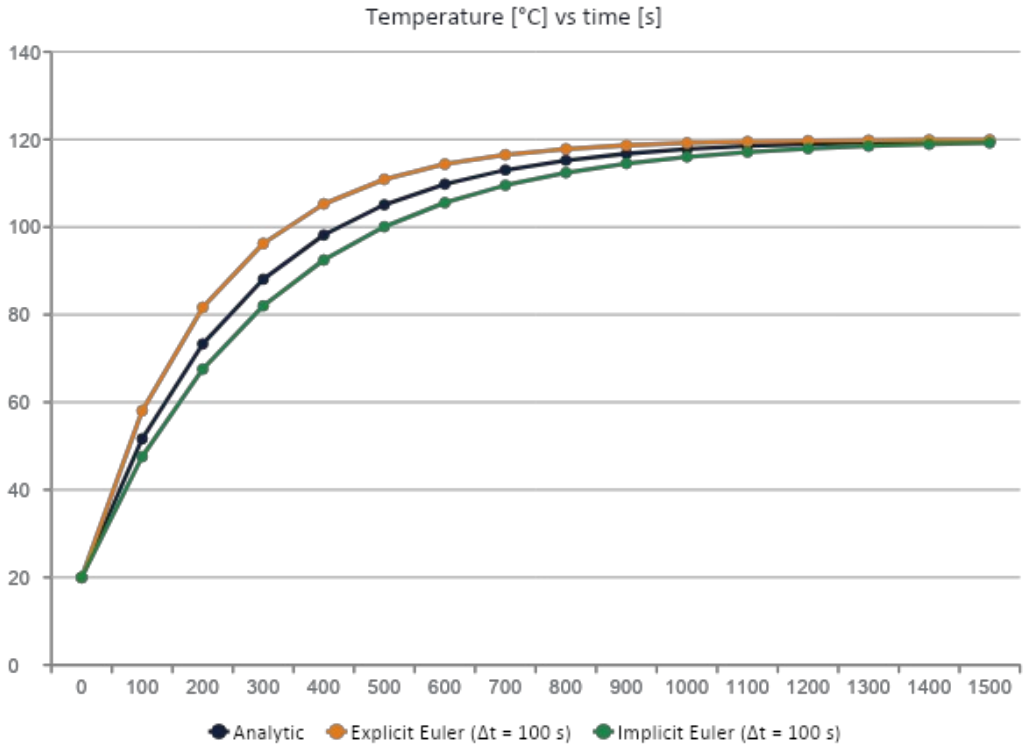
time constant τ = R·C

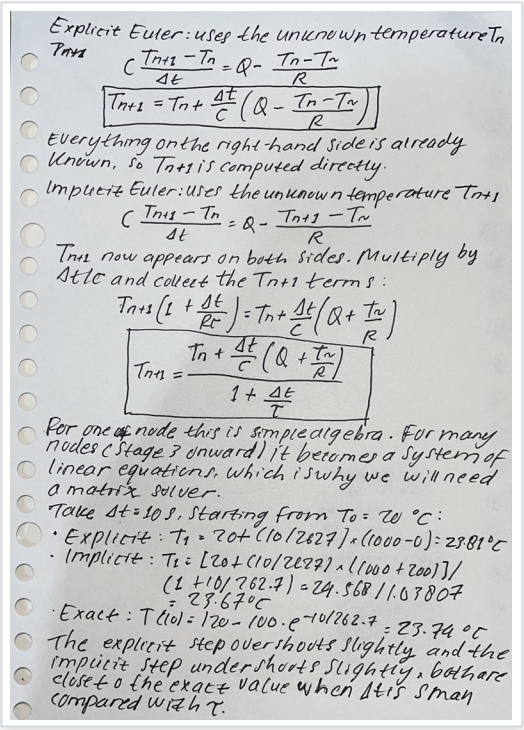
83.21 °C

T(τ): 63.2 % of the rise

Stage 2 — Analytic Solution & Time Discretisation

$T(t) = T_{ss} - (T_{ss} - T_0)\cdot e^{(-t/\tau)}$ vs Euler steps of $\Delta t = 100$ s





Explicit Euler: uses the unknown temperature T_n

$$C\frac{T_{n+1}-T_n}{\Delta t} = Q - \frac{T_n - T_\infty}{R}$$

$$T_{n+1} = T_n + \frac{\Delta t}{C}\left(Q - \frac{T_n - T_\infty}{R}\right)$$

Everything on the right-hand side is already known, so T_{n+1} is computed directly.

Implicit Euler: uses the unknown temperature T_{n+1}

$$C\frac{T_{n+1}-T_n}{\Delta t} = Q - \frac{T_{n+1} - T_\infty}{R}$$

T_{n+1} now appears on both sides. Multiply by $\Delta t/C$ and collect the T_{n+1} terms:

$$T_{n+1}\left(1 + \frac{\Delta t}{RC}\right) = T_n + \frac{\Delta t}{C}\left(Q + \frac{T_\infty}{R}\right)$$

$$T_{n+1} = \frac{T_n + \frac{\Delta t}{C}\left(Q + \frac{T_\infty}{R}\right)}{1 + \frac{\Delta t}{\tau}}$$

For one node this is simple algebra. For many nodes (Stage 3 onward) it becomes a system of linear equations, which is why we will need a matrix solver.

Take $\Delta t = 10$ s, starting from $T_0 = 20$ °C:

- Explicit: $T_1 = 20 + (10/2627)\times(1000 - 0) = 23.81$ °C
- Implicit: $T_1 = [20 + (10/2627)\times(1000 + 200)] / (1 + 10/262.7) = 24.568 / 1.03807 = 23.67$ °C
- Exact: $T(10) = 120 - 100\cdot e^{-10/262.7} = 23.74$ °C

The explicit step overshoots slightly and the implicit step undershoots slightly, both are closest to the exact value when Δt is small compared with τ.

Preview computed from the derived formulas; C++ implementation is the next step.

- Explicit uses the known T_n — it runs ahead of the exact curve.
- Implicit uses the unknown T_{n+1} — it lags behind.
- Both converge to 120 °C; the error shrinks as $\Delta t/\tau$ gets smaller.

Stage 2 — One Step by Hand & Stability

Explicit and implicit updates derived from the same energy balance

Explicit: $T_{n+1} = T_n + (\Delta t/C)\cdot[Q - (T_n - T\infty)/R]$

Implicit: $T_{n+1} = [T_n + (\Delta t/C)\cdot(Q + T\infty/R)] \,/\, (1 + \Delta t/\tau)$

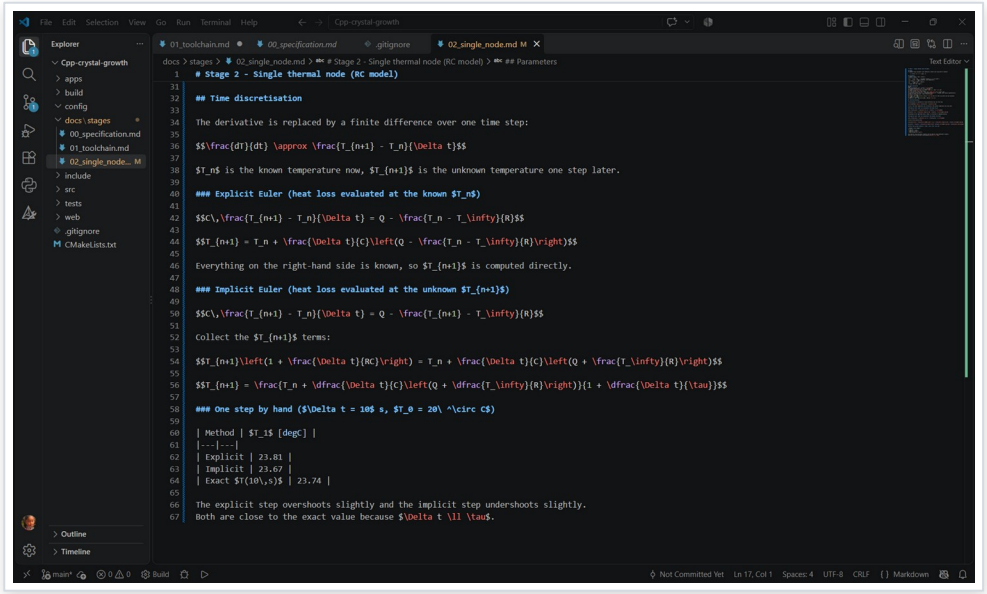
Δt = 10 s, T_0 = 20 °C	T_1 [°C]
Explicit Euler	23.81
Implicit Euler	23.67
Exact T(10 s)	23.74

Stability — what the stage-2 tests will prove

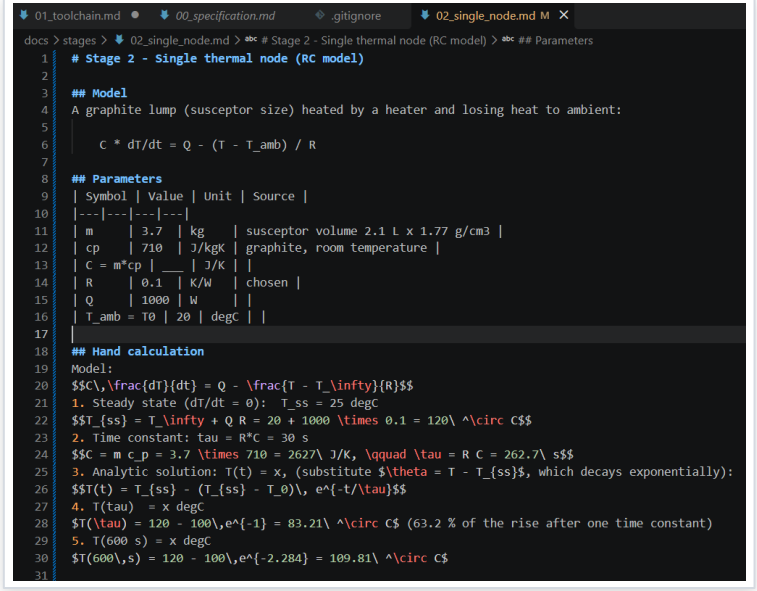
- Explicit error per step is multiplied by $(1 - \Delta t/\tau)$. If $\Delta t > 2\tau$ = 525 s, $|1 - \Delta t/\tau| > 1$ and the solution blows up.
- Implicit Euler divides by $(1 + \Delta t/\tau) > 1$, so it can never blow up — but a large Δt still gives a poor answer.
- This is why the reference software and CalculiX use implicit time stepping.
- Planned tests: error < 0.1 K at Δt = 1 s; explicit unstable at Δt = 600 s; implicit bounded 20–120 °C.

Documentation in the Repository

Stage notes in Markdown with LaTeX formulas, previewed in VS Code



docs/stages/02_single_node.md — time discretisation



Parameters and hand calculation

Next Steps

Resuming from Stage 2 implementation

Next	Deliverable	Proof
Stage 2 code	rc_node.hpp / .cpp, library in CMake	Tests vs 83.21 / 109.81 °C, stability tests
Stage 3	Thermal network G·T = Q, LU solver	Layered cylinder wall vs hand calculation
Stage 4	Radiation + k(T), Newton iteration	Two-cylinder radiation, 17.6 kW/m² (Part 3)
Stage 5	Furnace model from Part 3 geometry	Steady state vs FEA (susceptor, vessel, Q to water)
Stages 6–8	Melting, PID + recipe, interlocks	Energy balance closure, step response, fault tests
Stages 9–12	HAL, simulator, HMI, docs	Running HMI like the target screen

Resume in one minute (full checklist in PROGRESS.md)

```
cmake -S . -B build
cmake --build build --config Release
ctest --test-dir build -C Release --output-on-failure
```

Then: rc_node.hpp → rc_node.cpp → add_library in CMake → stage-2 tests → commit and tag stage-02.

Summary

Done

- Toolchain installed and verified (MSVC 19.51, CMake, CTest, Git).
- Repository structure and stage-note system set up.
- Stage 2 model derived by hand: T_{ss} = 120 °C, τ = 262.7 s, $T(\tau)$ = 83.21 °C.
- Explicit and implicit Euler derived and checked with one step.

Design

- Layered: Model → HAL → Control → Application → HMI.
- Same control code for simulator and real furnace.
- Every layer tested against hand calculations and Part 3 FEA.

Next

- Implement and test the RC node (stage 2).
- Thermal network and LU solver (stage 3).
- Radiation with Newton iteration (stage 4), then the furnace model.